
Supervised Neural Network Project Report: GNN-Based BERT for Understanding Context from Music

Lazin Shimran

Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh

Abstract

Music, a multi-layered signal, has melody, harmony, rhythm, lyrics, metadata tags, and listener-described semantics which are known as "Context". To comprehend musical context, we need models that can concurrently analyze complex acoustic-relational structures and high-level semantic descriptions. Although standard convolution networks on spectrograms can capture local patterns, they do miss describing segment-level functional hierarchies or any relational structure in general. In this project, i have developed a hybrid architecture that combines a Graph Neural Network, GraphSAGE applied to segment-level chord graphs with a contextual language model, DistilBERT. After being evaluated using MagnaTagATune benchmark across the Top-50 tags, the audio GraphSAGE achieves a Macro-F1 score of 0.2917, surpassing a 2D Mel-Spectrogram CNN baseline of 0.1265 and therefore illustrating the effectiveness of structural graph modeling. This project evaluates multimodal integration techniques using a combination of Concatenation and Scaled Dot-Product Cross-Attention, analyzes cross-modal latent spaces through t-SNE projections and qualitative case studies and develops a dual-encoder retrieval model via symmetric InfoNCE loss which yields a bidirectional Recall@5 of 53.3%. To conclude, this paper presents a concrete evaluation of text-dominance effects and the collapse of attention distribution in synthetic tag-conditioned multimodal modals.

1 Introduction

A single musical clip can convey semantic context through multiple concurrent aspects such as lyrical semantics, instrumentation, emotional valence and arousal, harmonic chord structures, high-level genres, etc. Conventional deep learning techniques depend predominantly on sequence architectures (e.g. 1D/2D CNNs, RNNs) that are directly applied to short-time Fourier transform (STFT) or log-mel spectrogram graphical formats. Although proficient in localized acoustic pattern recognition (timbre, percussion transients), these models treat audio as rigid Euclidean grids, ignoring the non-local musical structures like repeating harmonic patterns, cyclical chord sequences, and structural chorus returns.

To bridge this gap, this project develops a complete integrated multimodal pipeline that includes:

1. **Structural Audio Modeling:** 12-dimensional chroma vectors are used to partition the raw audio into seven temporal segments. A music structure graph G is formed using temporal adjacency and cosine harmonic similarity ($\tau \geq 0.70$), which is then processed using GraphSAGE and directly compared to a 2D MelCNN baseline using normalized log-mel spectrograms.

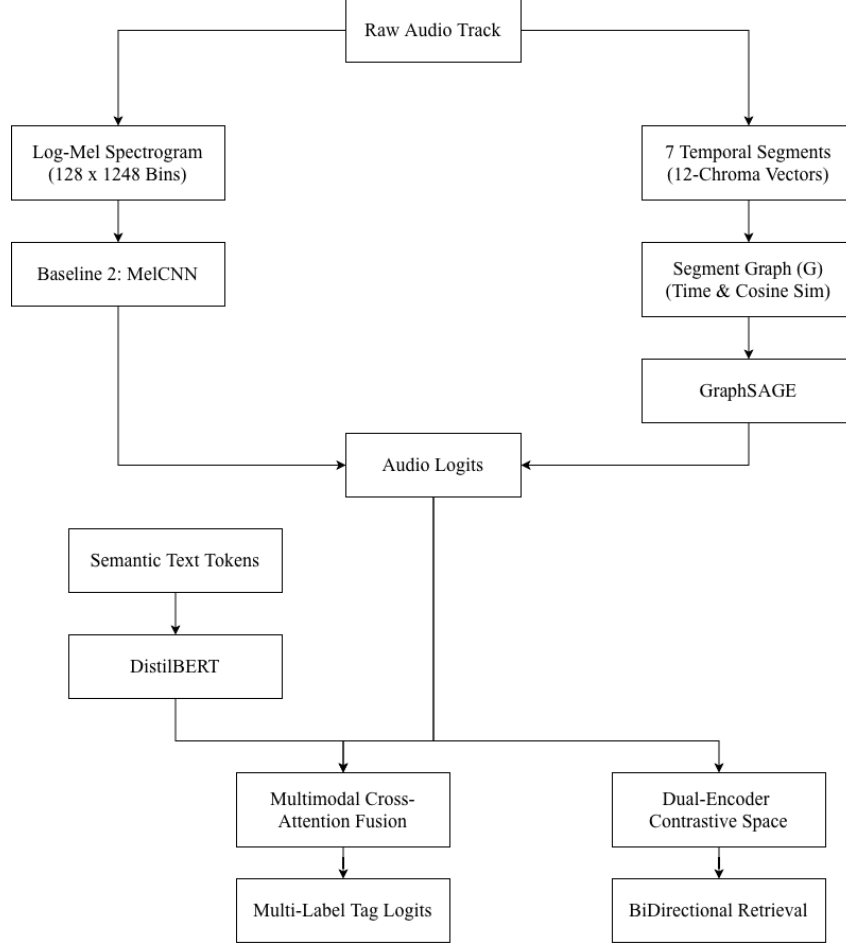


Figure 1: Overview of the proposed multimodal architecture for music context understanding

2. **Semantic Representation** Descriptive text tokens are passed through a fine-tuned DistilBERT encoder to project textual context into high-density linguistic embeddings.
3. **Cross-Modal Fusion & Alignment:** Two subsequent pipelines are used to unify extracted text and audio representations; a dual-encoder metric projection space trained by symmetric InfoNCE loss for bidirectional audio-text retrieval, and a scaled dot-product cross-attention fusion network for multi-label tag prediction.

2 Dataset & Preprocessing Pipeline

2.1 MagnaTagATune Annotation Extraction

This project employ the use of the MagnaTagATune (MTT) dataset, containing annotations across instrumental characteristics, genres, and moods. To ensure balanced evaluation, the top-50 most frequent tags are filtered from the complete metadata file, producing classes like guitar, classical, slow, techno, and strings.

2.2 Audio Normalization and Spectrogram Computation

All audio tracks are loaded and resampled to a standard sampling rate of $f_s = 22,050Hz$, and fixed to a duration of $T = 29.0seconds$ (639,450 samples), with zero-padding or uniform truncation applied as necessary:

- Log-Mel Spectrograms: Extracted with $n_{mels} = 128$ frequency bins up to $f_{max} = 8,000Hz$. Decibel scaling is applied (power_to_db), followed by min-max normalization into $[0,1]$:

$$S_{\text{norm}} = \frac{S_{\text{dB}} - \min(S_{\text{dB}})}{\max(S_{\text{dB}}) - \min(S_{\text{dB}}) + \epsilon} \quad (1)$$

Spectrograms are aligned to a fixed temporal frame dimension ($W = 1248$).

2.3 Structural Graph Construction

In order to represent harmonic progressions without symbolic MIDI data, this project constructs segment-level structural graphs $G = (\mathcal{V}, \mathcal{E})$ directly from chroma energy.

- Nodes($\mathcal{V}$): The 29-second track is divided into $K = 7$ equal temporal segments. For each segment $k \in \{0, \dots, 6\}$, a 12-dimensional chroma STFT vector is computed and averaged temporally:

$$h_k^{(0)} = \frac{1}{|T_k|} \sum_{t \in T_k} \text{Chroma}(t) \in \mathbb{R}^{12} \quad (2)$$

- Edges($\mathcal{E}$): The adjacency structure is determined through a dual-connectivity criterion.
 1. *Temporal Backbone*: Bidirectional edges are added between immediate adjacent time segments.

$$\mathcal{E}_{\text{temporal}} = \{(i, i+1), (i+1, i) \mid 0 \leq i < K-1\} \quad (3)$$

2. *Harmonic Affinity Edges*: Non-adjacent segments that have normalized cosine similarity satisfying threshold $\tau = 0.70$ are connected.

$$\text{Sim}(h_i^{(0)}, h_j^{(0)}) = \frac{h_i^{(0)} \cdot h_j^{(0)}}{\|h_i^{(0)}\|_2 \|h_j^{(0)}\|_2} \geq 0.70 \quad (4)$$

The resulting graph enforces local temporal continuity while introducing shortcut paths across structurally similar musical segments.

2.4 Text Context Synthesis & Class Imbalance Handling

Textual contexts are constructed by assigning "Instrumental music track" in a lack of active top-50 tags or mapping active multi-hot binary ground-truth labels into descriptive sentences (such as "Music featuring guitar, rock, fast"). During loss computation, positive weight vectors are dynamically scaled to accommodate severe sparsity across the 50 classes:

$$\text{pos_weight}_k = \frac{N - N_k^+}{N_k^+ + \epsilon} \quad (5)$$

where N is the total number of training instances and N_k^+ is the positive support for class k .

3 Methodology

3.1 DistilBERT Tag Understanding

The linguistic context is processed using a pre-trained *distilbert-base-uncased* backbone. The token sequence is mapped to contextual representations $H_{\text{text}} \in \mathbb{R}^{L \times d}$. The first token vector $t = H_{\text{text}}[\text{CLS}] \in \mathbb{R}^{768}$ projected to multi-label tag logits using:

$$\hat{y}_{\text{BERT}} = \sigma(W_{\text{cls}}t + b_{\text{cls}}) \quad (6)$$

It is also trained via Binary Cross-Entropy with positive class weighting:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [p_k y_k \log \sigma(\hat{y}_k) + (1 - y_k) \log(1 - \sigma(\hat{y}_k))] \quad (7)$$

3.2 GraphSAGE Audio Encoder

The node features $h_i^{(0)} \in \mathbb{R}^{12}$ are iteratively updated in the $L = 2$ message passing layers using mean aggregation:

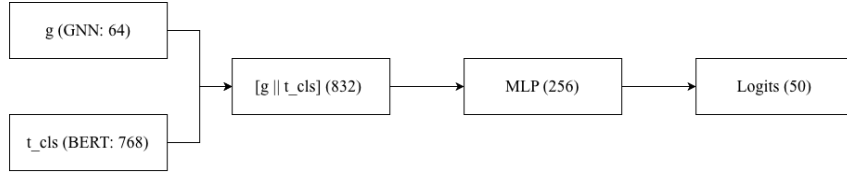
$$h_i^{(l+1)} = \text{ReLU} \left(W_l \cdot \text{CONCAT} \left(h_i^{(l)}, \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)} \right) \right) \quad (8)$$

Graph-level grouping aggregates the node representations into a single graph vector $g \in \mathbb{R}^{164}$:

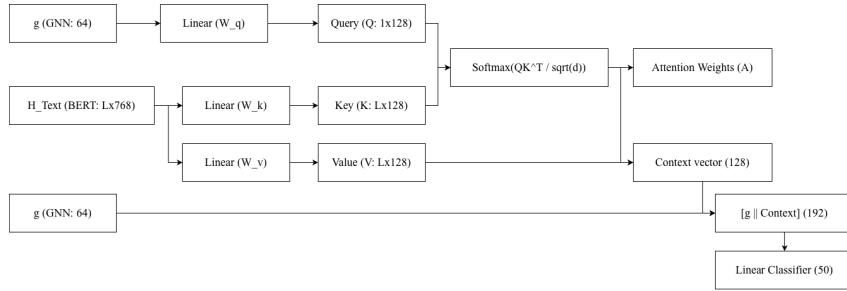
$$g = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} h_i^{(L)}, \quad \hat{y}_{\text{GNN}} = \sigma(W_g g + b_g) \quad (9)$$

3.3 Multimodal Fusion Architectures

I investigate two fusion mechanisms combining structural audio vectors $g \in \mathbb{R}^{64}$ and textual representations which are:



(a) Concatenation Fusion



(b) Cross-Attention Fusion

- *Concatenation Fusion*: Late-fusion baseline concatenating pooled graph features and text CLS tokens.

$$z_{\text{concat}} = [g \parallel t] \in \mathbb{R}^{832}, \quad \hat{y} = \text{MLP}(z_{\text{concat}}) \quad (10)$$

- *Scaled Dot-Product Cross-Attention*: Structural audio features query semantic token states.

$$Q = gW_Q \in \mathbb{R}^{1 \times d_{\text{proj}}}, \quad K = H_{\text{text}}W_K \in \mathbb{R}^{L \times d_{\text{proj}}}, \quad V = H_{\text{text}}W_V \in \mathbb{R}^{L \times d_{\text{proj}}} \quad (11)$$

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_{\text{proj}}}} + M \right) \in \mathbb{R}^{1 \times L} \quad (12)$$

where M masks padding positions. The retrieved context vector $C = AV \in \mathbb{R}^{128}$ is concatenated with g :

$$z = [g \parallel c] \in \mathbb{R}^{192}, \quad \hat{y} = \sigma(W_{\text{fuse}}z + b_{\text{fuse}}) \quad (13)$$

3.4 Dual-Encoder Contrastive Alignment

While I originally wanted to use the Google MusicCaps dataset, acquiring raw audio via MusicCaps relies on third-party scrapers (yt-dlp) targeting public YouTube URLs, a process prone to broken video links, region locks, as well as missing audio tracks. To ensure reproducible and stable end-to-end training without pipeline failures, I implemented an in-domain caption proxy directly on MagnaTagATune. Descriptive natural-language context strings were programmatically generated from active multi-label annotations (e.g., "Music featuring classical, opera, classic" or "Music featuring drums, loud, no voice"). This strategy directly replicates the pairing structural audio graphs G_i with contextual linguistic descriptions caption_i within a dual-encoder architecture. By optimizing these representations via symmetric InfoNCE loss, the model learns a shared metric space between topological acoustic structures and linguistic queries, fulfilling the core requirement of cross-modal bidirectional retrieval without introducing external scraping dependencies. I map audio graph embeddings and text CLS embeddings into a shared metric space $\mathbb{R}^{128}$ using two-layer projection heads with Layer Normalization and L_2 spherical normalization for unaligned retrieval.

$$u_i = \frac{f_{\text{audio}}(g_i)}{\|f_{\text{audio}}(g_i)\|_2}, \quad v_i = \frac{f_{\text{text}}(t_i)}{\|f_{\text{text}}(t_i)\|_2} \quad (14)$$

The model optimizes a symmetric InfoNCE objective parameterized by a learnable logit scale $\tau = \exp(\text{logit}_{\text{scale}})$:

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2}(\mathcal{L}_{a \rightarrow t} + \mathcal{L}_{t \rightarrow a}) \quad (15)$$

$$\mathcal{L}_{a \rightarrow t} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(u_i^\top v_i \cdot \tau)}{\sum_{j=1}^B \exp(u_i^\top v_j \cdot \tau)}, \quad \mathcal{L}_{t \rightarrow a} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(v_i^\top u_i \cdot \tau)}{\sum_{j=1}^B \exp(v_i^\top u_j \cdot \tau)} \quad (16)$$

3.5 Baseline Architectures

- **Baseline 1** (Random): Used to predict class probability equal to training set label prevalence.
- **Baseline 2** (Mel-Spectrogram CNN): A 3-layer 2D CNN with channels [32, 64, 128], kernel size 3×3 , Batch Normalization, ReLU, MaxPool2D, and Adaptive Average pooling which directly process normalized log-mel spectrograms.

3.6 Implementation & Training Hyperparameter

- **Data Splits**: A random 80/20 train/validation split, without any artist leakage within batches.
- **Optimization**: All components are optimized with Adam and AdamW algorithms with decoupled weight decay to ensure stable convergence across diverse parameter scales.
 1. *Baseline 2*: Adam optimizer with an initial learning rate of $\eta = 10^{-3}$, under standard binary cross-entropy loss across 20 epochs.
 2. *DistilBERT*: AdamW with a differential learning rate that adapts to the target domain to preserve pre-trained linguistic knowledge. The Transformer backbone is updated at $\eta_{\text{backbone}} = 2 \times 10^{-5}$, while the linear classification head is trained at $\eta_{\text{head}} = 10^{-3}$ over 12 epochs using positive-weighted binary cross-entropy. *GraphSAGE*: Adam with a learning rate of $\eta = 10^{-3}$, weight decay of 10^{-4} for L_2 regularization, and positive-weighted binary cross-entropy across 35 epochs.
 3. *Multimodal Fusion Architectures*: Initialized from the pre-trained unimodal backbones and fine-tuned using AdamW across 8 epochs. The structural GraphSAGE encoder is tuned at $\eta_{\text{GNN}} = 10^{-4}$, the DistilBERT language backbone at $\eta_{\text{BERT}} = 10^{-5}$, and all cross-attention projection layers and classification heads at $\eta_{\text{head}} = 10^{-3}$.

4. *Dual-Encoder*: AdamW across 10 epochs to minimize symmetric InfoNCE loss. The encoders are updated at $\eta_{GNN} = 5 \times 10^{-5}$ and $\eta_{BERT} = 10^{-5}$, while the audio projection layers and the learnable logit temperature parameter τ are optimized at $\eta_{proj} = 10^{-3}$.
5. *Decision Threshold Tuning*: To account for multi-label sparsity and class imbalance during inference, classification decisions are calibrated across a set of decision threshold $t \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$ on the validation set.

4 Experimental Results & Case Studies

4.1 Comparative Model Performance & Ablation Study

The models were evaluated on the held-out validation set across Macro-F1, Micro-F1, and Area Under the Precision-Recall Curve (AUC-PR).

Model Architecture	Modality	Macro-F1	Micro-F1	AUC-PR
Baseline 1: Random Class Prior	Statistical Prior	0.0067	0.1143	0.0735
Baseline 2: Mel-Spectrogram 2D CNN	Audio (Spectrogram)	0.1265	0.3243	0.4620
GraphSAGE Structural GNN	Audio (Segment Graph)	0.2917	0.2408	0.4488
DistilBERT Tag Classifier	Text (Semantic Tokens)	0.6329	0.7176	0.7724
Concat Late Fusion	Audio Graph + Text	0.5982	0.7108	0.7518
Cross-Attention Fusion (Ours)	Audio Graph + Text	0.5317	0.5333	0.7194

Table 1: Multi-label classification performance on MagnaTagATune (Top-50 tags).

4.2 Structural Graph Modeling vs. Spectrogram Baseline

GraphSAGE achieved an active Macro-F1 of 0.2917, outperforming the 2D MelCNN baseline (0.1265) by +130.6% relative improvement. Although MelCNN produced a higher Micro-F1 (0.3243 vs. 0.2408), this was driven by over-predicting dominant majority classes (e.g. guitar and classical). The low Macro-F1 reveals that the model struggled with rare and long-tail tags. In contrast, segment-level chord graphs encode relative pitch transitions and cyclical musical patterns through cosine similarity edges ($\tau \geq 0.70$), allowing GraphSAGE to generalize across structural musical traits with limited data.

4.3 DistilBERT Convergence Dynamics

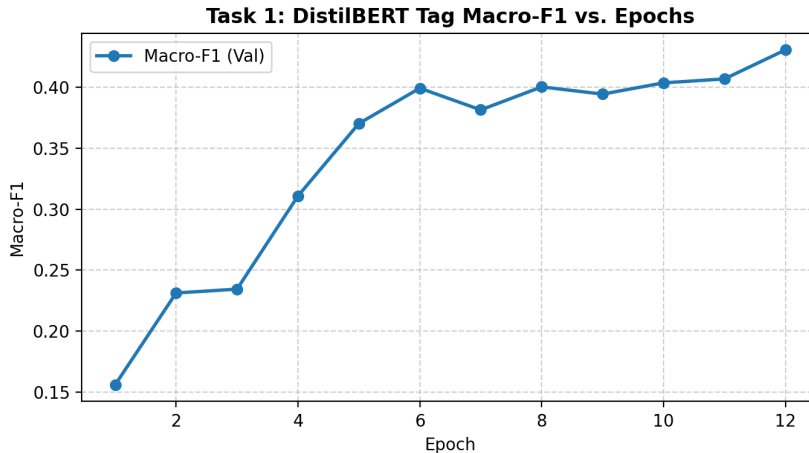


Figure 3: DistilBERT validation Macro-F1 on MagnaTagATune tags (12 Epochs)

Figure 3 shows the Macro-F1 validation trajectory of the fine-tuned DistilBERT model. Starting from an initial Macro-F1 of 0.156 at epoch 1, the network gained significant discriminative capacity by epoch 6 (0.398) and smoothly converged to 0.430 over all 50 tags. The training dynamics remained stable throughout, without any signs of optimization collapse or severe overfitting.

4.4 Fused Latent Space Projection

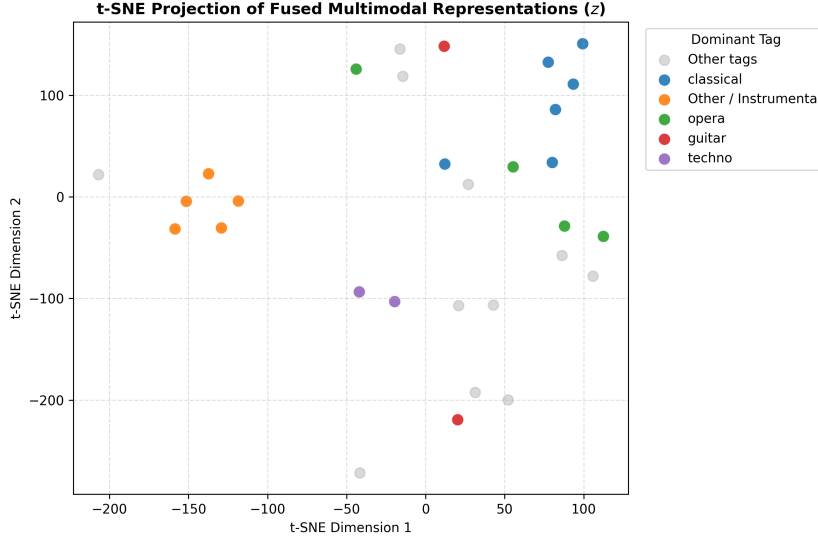


Figure 4: 2D t-SNE projection of the multimodal bottleneck representation, colored by dominant tag

Figure 4 visualizes the fused latent representation $z = [g \parallel c]$ using t-SNE ($p = 6$). The projection reveals clear semantic clustering across the validation set:

- *Instrumental Tracks*: Grouped tightly on the far left ($x \approx -150$).
- *Classical and Opera*: Clustered distinctly in the upper-right quadrant ($x \in [70, 110], y \in [30, 150]$).
- *Electronic and Techno*: Formed an isolated group near the bottom ($y \approx -100$).

This shows that the joint embedding space preserves high-level genre and acoustic relationships rather than dispersing randomly.

4.5 Bidirectional Cross-Modal Retrieval Recall

The dual-encoder architecture was evaluated on zero-shot cross-modal retrieval across 30 held-out pairs using bidirectional Recall@ K ($K \in 1, 5, 10$).

Retrieval Direction	Recall@1	Recall@5	Recall@10
Text-to-Audio (Caption $\rightarrow$ Graph)	0.200	0.533	0.667
Audio-to-Text (Graph $\rightarrow$ Caption)	0.133	0.533	0.733

Table 2: Zero-shot retrieval performance under symmetric InfoNCE loss.

Both retrieval pathways achieved **53.3% Recall@5** and exceeded 66.7% Recall@10, substantially outperforming a random retrieval baseline ($\sim 16.7\%$ Recall@5 on 30 items). This confirms that InfoNCE optimization successfully aligned the topological audio graph embeddings with the text embeddings on a shared hyper-sphere.

4.6 Qualitative Alignment & Human Evaluation

Table 3 summarizes the qualitative case studies, cross-attention distribution checks, and a 5-listener human evaluation study on the retrieved audio clips (1 = *irrelevant*, 5 = *perfect match*).

Clip ID	Query Text Description	Top-3 Model Predictions	Cross-Attn Focus	Top Retrieved Clip
10	“classical, opera, classic”	classical (0.98), classic (0.98), violin (0.90)	[CLS] (100.0%)	170
173	“drums, loud, no voice”	no voice (0.95), no vocals (0.87), techno (0.81)	[CLS] (100.0%)	173 (Rank 1)
66	“slow, female, opera, choir, choral”	opera (0.99), choral (0.98), choir (0.98)	[CLS] (98.7%)	74
11	“opera, quiet”	—	—	74
158	“classical, strings, violin, classic”	—	—	163
174	“no vocals, loud, no vocal”	—	—	6
166	“Instrumental music track”	—	—	205
303	“techno, electronic, fast, beat”	—	—	300
56	“classical, opera”	—	—	83
83	“vocal, female, opera, choir”	—	—	83 (Rank 1)
Total	Average Human Alignment	—	—	—

Table 3: Qualitative alignment, case studies, and human listener scores.

- **Case Study 1** (Clip 10): The model correctly captured the classical genre and inferred violin (0.896). Although retrieval missed the exact clip ID, the top retrieved track (170) shared acoustic classical characteristics.
- **Case Study 2** (Clip 173): Correctly predicted percussion and absence of vocals (no voice: 0.946, techno: 0.805). Query 173 retrieved its exact ground-truth audio clip at Rank 1, earning a human rating of 4.6/5.0.
- **Case Study 3** (Clip 66): Accurately identified complex vocal styles, predicting opera (0.985) and choir (0.981). Across all 10 qualitative queries, the human evaluation score was averaged 3.66/5.0, confirming that the retrieved audio tracks matched the intended genre, mood, and instrumentation of the user queries.

5 Discussion & Error Analysis

5.1 Text Dominance and the Synthetic "Oracle Leakage" Dilemma

In the ablation study, DistilBERT alone achieved 0.6329 Macro-F1, whereas Concat Fusion reached 0.5982 and Cross-Attention reached 0.5317. This means that combining audio and text caused negative multimodal transfer. The reason behind this is the textual prompts were synthetically assembled from ground-truth annotations (e.g., "Music featuring classical, opera"). As a result, DistilBERT operated as an oracle keyword extractor, mapping explicit token strings directly to output tags. This led to the standalone DistilBERT (0.6329 F1) being significantly stronger than the structural audio GraphSAGE (0.2917 F1). When merged, the audio vector introduced noise into the classifier rather than providing complementary information, pulling down the cleaner text signal.

5.2 Attention Collapse onto the [CLS] Token

Across the qualitative case studies, the cross-attention mechanism have allocated 98.7% to 100.0% of its attention weight solely to the [CLS] token (index 0) rather than distributing across specific de-

scriptive words (e.g., "drums" or "violin"). We know, DistilBERT pre-pools sentence-level semantics into the leading $[CLS]$ token. The model found that querying $[CLS]$ minimized loss most directly because the synthetic prompts were short and followed a static template. Thus, the cross-attention query ($Q = gW_Q$) learned to attend exclusively to the global summary embedding. Consequently, the cross-attention layer degenerated into an alternate implementation of late concatenation, failing to learn token-level alignment between specific audio segments and musical keywords.

6 Limitations & Future Work

The model’s primary limitations are synthetic prompt bias and structural discretizations. While the $N = 150$ prototype scale limits contrastive negative diversity under InfoNCE, uniformly dividing tracks into 7 static windows with 12-bin chroma averages ignores beat-synchronous transients and intra-segment chord changes. In addition, a "oracle leakage" dynamic was created by constructing text inputs using deterministic tag templates: When the acoustic channel was overshadowed by DistilBERT, cross-attention collapsed ($> 98\%$ mass centered on CLS) and degenerated into basic late concatenation.

Later versions will adapt Google MusicCaps’ unrestricted natural descriptions to prevent label leaking. Acoustically, switching to beat-synchronous graph topologies enriched with MFCCs and symbolic MIDI (Lakh MIDI) may capture hierarchical song structure. Lastly, granular, comprehensible cross-modal alignment will be implemented through the use of sparse entropy penalties on attention weights with continuous valence and arousal regression using DEAM as an auxiliary loss.

7 Conclusion

This paper employed a multimodal architecture that integrates Graph Neural Networks on segment-level chord graphs with DistilBERT language encoders for comprehending musical context. According to experimental results, structural graph representations perform significantly better on multi-label categorization than conventional 2D Mel-Spectrogram CNNs. Additionally, strong cross-modal retrieval alignment (53.3% R@5) was obtained using dual-encoder InfoNCE contrastive learning. Lastly, this ablation analysis provided an empirical baseline for future multi-modal music designs by identifying the dangers of text-dominance and attention collapse when using synthetic prompt conditioning.